Predicate Logic

IT3113 & CSH3143

Why Predicate Logic?

The propositional logic is not powerful enough to

represent all types of assertions

to express certain types of relationship between propositions

Example

X is greater than 1

Here X is a variable

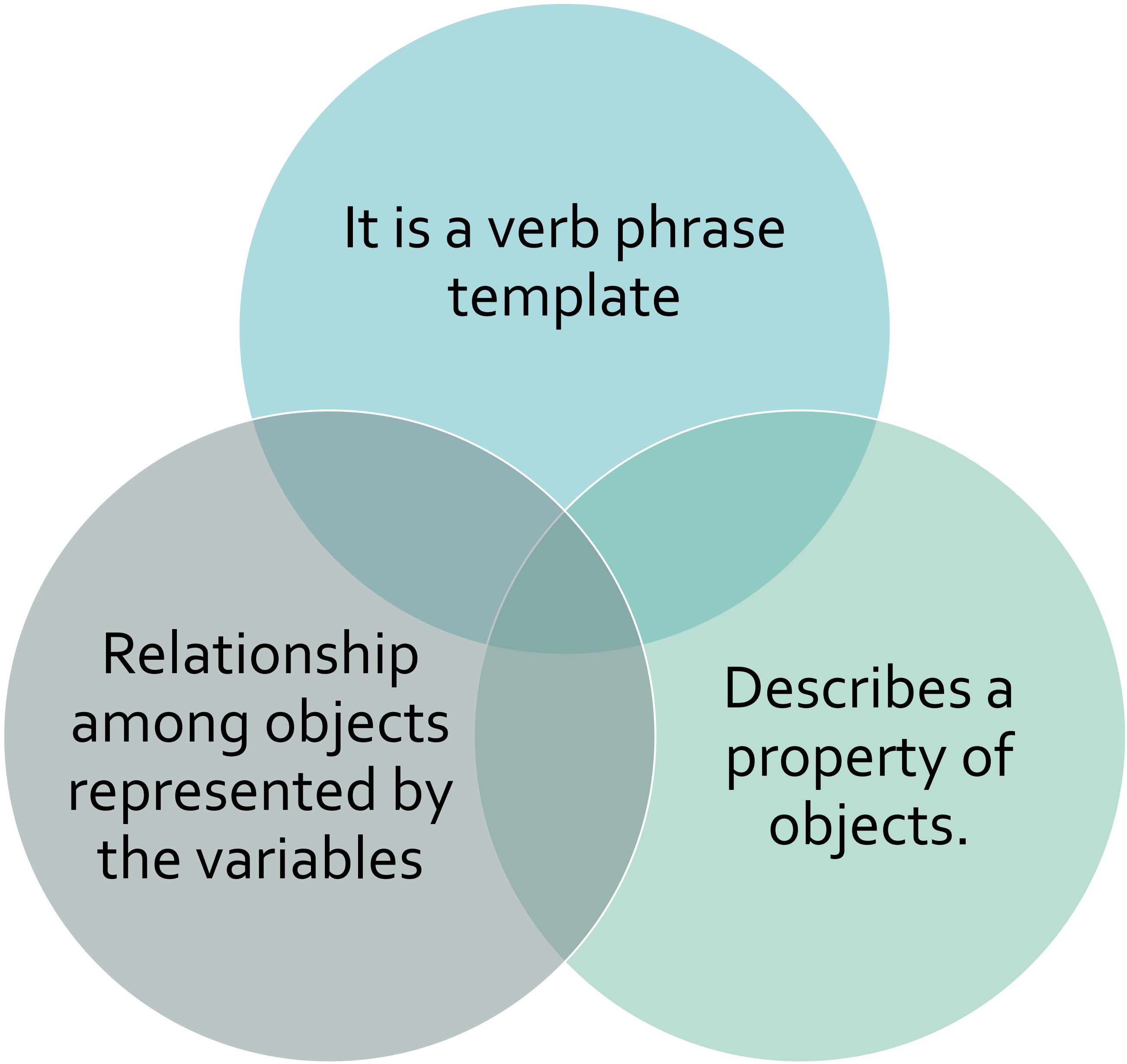
This is not a proposition

Propositional logic can't deal with this type of statements

Thus we need more powerful logic

Predicate Logic is one of it

Predicate


It is a verb phrase template
Relationship among objects represented by the variables
Describes a property of objects.

Example

The car Tom is driving is blue.

is_blue(car).

The sky is blue

is_blue(sky).

The cover of this book is blue.

is_blue(cover).

Quantification

A predicate with a variable is not a proposition.

Example: **Y is greater than 10.**

- With variable Y over the universe of real numbers

Quantification …

A predicate with variables can be made into a proposition by applying either of the following:

- Assign a value to the variable
- Quantifying the variable using quantifiers

Universe of Discourse

The set of objects of interest.

The propositions in the predicate logic are statements on objects of a universe.

Universe is thus the domain of the variables.

Example

- The set of real numbers.
- The set of integers.
- The set of all cars in a parking lot.
- The set of all students in a classroom.

Quantifiers

The universal quantifiers

The existential quantifiers

Universal Quantifier

The expression: $\forall X \; P(X)$ denotes the universal quantification of the atomic formula $P(X)$.

The expression is understood as

- For all X, P(X) holds.
- For each X, P(X) holds.
- For every X, P(X) holds.

Universal Quantifier …

Example:

All cars have wheels

- Can be represented as $\forall X\ P(X)$
  - Where $P(X)$ is the predicate denoting that **X has wheels**.
  - Here, the **universe of discourse** is **only populated with cars**.

Existential Quantifier

The expression: $\exists x\ P(x)$, denotes the existential quantification of P(X).

The expression is understood as

- There exist an X such that P(X).
- There is at least one X such that P(X).

Existential Quantifier …

Example:

Someone loves you

- Can be represented as **$\exists x\ P(x)$**
  - Where **P(X)** is the predicate meaning that **X loves you**.
  - Here, the **universe of discourse** contains **all living creatures**

The Standard Quantifier - Connective Pairing

Universal quantifier pairs with implication ($\longrightarrow$):

- $\forall X\ (P(X) \longrightarrow Q(X))$
- if x has property P, then X has property Q

Existential quantifier pairs with conjunction ($\land$):

- $\exists X\ (P(X) \land Q(X))$
- there is some x that has both property P and Q

Example

All men are mortal.

Incorrect: $\forall X\ (Man(X) \land Mortal(X))$ - Meaning: everything in the universe is both a man and mortal

Correct: $\forall x\ (Man(x) \rightarrow Mortal(x))$ - for anything that happens to be a man, it is also mortal

Example …

Some students are hardworking

Incorrect: **$\exists X\ (Student(X) \rightarrow Hardworking(X))$** - As long as the domain contains at least one non-student anywhere, this formula is true even in a world where no student is hardworking at all.

Correct: **$\exists X\ (Student(X) \land Hardworking(X))$** - This forces the witness X to genuinely satisfy both conditions simultaneously

Some Special Cases

No X is Y:

- $\forall x\,(X(x) \rightarrow \neg Y(x))$

Not all X are Y:

- $\neg\forall x\,(X(x) \rightarrow Y(x))$,
- Equivalently, $\exists x\,(X(x) \wedge \neg Y(x))$

Only X's are Y:

- $\forall x\,(Y(x) \rightarrow X(x))$ (note the direction: reversed from what students often expect)

X unless Y:

- $\neg Y \rightarrow X$ (treat like "if not Y then X")

The only X is a:

- $\forall x\,(X(x) \rightarrow x = a)$

Order of Application

When more than one variables are quantified in a WFF such as $\exists Y, \forall X\ P(X, Y)$

They are applied from the inside, that is one closest to the atomic formula is applied first.

Thus, $\exists Y, \forall X\ P(X, Y)$ reads $\exists Y, [\forall X\ P(X, Y)]$

we say that **"There exists a Y such that for every X, P(X,Y)"** holds.

Representing the sentence in Predicate Logic

Before any sentence can be processed by a knowledge-based system, it must first be expressed as a well-formed formula (WFF) in predicate logic

This step is fundamentally a translation task: converting an informal, ambiguous natural language sentence into a precise, formal, symbolic representation.

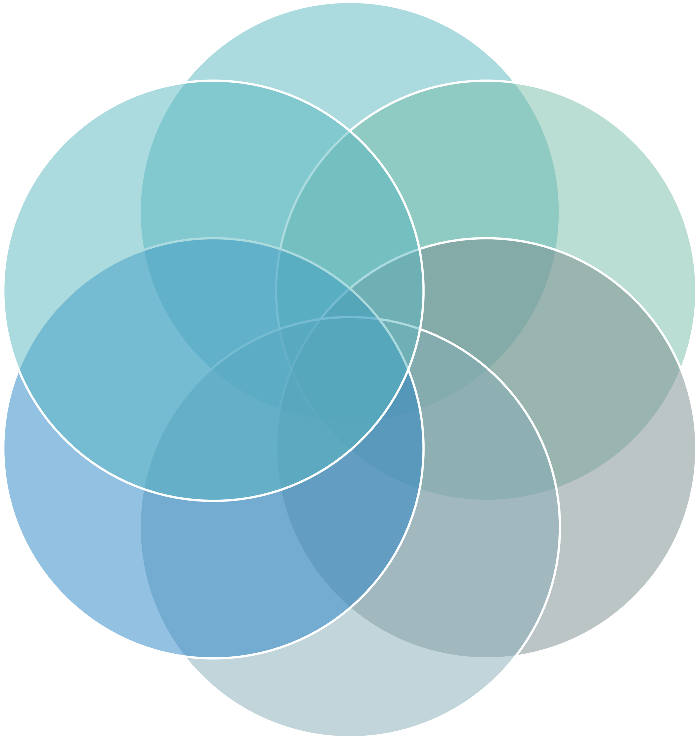
Properties of a Predicate Expression

Constant

- Names one specific object
- Example: john, sri_lanka, exam1

Connectives

- Combine formula
- $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$



Variable

- Stands for an unspecified/any object
- Example: X, Y, Z

Quantifiers

- Generalizes over objects
- $\forall$ (for all), (there exists)

Predicate

- Expresses a property or relation; evaluates to true/false
- Example: student(X), teaches(X,Y)

Function

- Maps object(s) to another object (not true/false)
- Example: MotherOf(X)

A systematic approach for translation

Step 1: Identify the objects.

- What are the "things" the sentence talks about?
- These become constants (specific, named objects) or variables (general, unspecified objects).

Step 2: Identify the properties and relations.

- What is being said about these objects?
- These become predicates. A property of one object is a unary predicate (e.g., Student(x); a relationship between two or more objects is a predicate of higher arity (e.g., Teaches(x,y)).

A systematic approach for translation

Step 3: Identify quantifier words. Look for signal words:

- Universal (∀): all, every, each, any, no ... Is
- Existential (∃): some, there exists, a certain, at least one.

Step 4: Assemble the formula, applying the correct pairing convention and paying attention to scope and order.

Some Examples

Sentence	Objects/ Predicates	Predicate Logic
Mathan is a man.	Constant: mathan Predicate: Man	Man(mathan)
All men are mortal.	Variable: X Predicates: Man, Mortal	$\forall X\ (Man(X) \rightarrow Mortal(X))$
Some students are hardworking.	Variable: X Predicates: Student, Hardworking	$\exists X\ (Student(X) \wedge Hardworking(X))$
John teaches Mary.	Constants: john, mary Predicate: Teaches	Teaches(john, mary)
Every lecturer has a student.	Variables: X, Y Predicates: Lecturer , Student , Has	$\forall X\ (Lecturer(X) \rightarrow \exists Y\ (Student(Y) \wedge Has(X, Y)))$
No bird can swim.	Variable: X Predicates Bird, Swim	$\forall X\ (Bird(X) \rightarrow \neg Swim(X))$
Only students may enter the lab.	Variable: X Predicates: Enter, Student	$\forall X\ (Enter(X, lab) \rightarrow Student(X))$
John's mother is a doctor.	Constant: john Function: MotherOf Predicate: Doctor	Doctor(MotherOf(john))

Exercises

Convert each of the following to predicates

1. John gives the book to Mary.
2. Jim gives a flower to Jeni.
3. Jerry gives a hat to Tim.
4. Mala talks to Sheela.
5. Mathan comes to the temple.
6. Bala goes to school.

Exercises …

Convert each of the following to predicates

1. Every employee has a manager.
2. Some lecturers are also researchers.
3. No student who cheats passes the course.
4. Only postgraduate students may access the research lab
5. John likes everyone who likes Mary.
6. There is a course that every student in the department must take.

Exercises …

Convert each of the following to predicates

7. Every AI student is a CS student
8. Every dog in the village has bitten a thief.
9. Rajah is a student, but he does not study
10. Shanika, who is a doctor, is taller than Samintha
11. Those who play chess are good in planning
12. Bala goes wherever Mala goes, but not to the temple
13. Mala likes whatever Bala likes
14. If Ramya is pretty, then she is not intelligent.

Exercises …

Convert each of the following to predicates

15. Ramya is either pretty or intelligent
16. Ramya is neither pretty nor intelligent
17. Even numbers greater than two are not prime.
18. Some students who study science in A/L are intelligent.
19. Everyone has a mother.
20. Sita hates everything Ravanan likes.

Clausal Form

Predicate Logic

Why Clausal Form?

Automated reasoning systems (resolution provers, Prolog's inference engine) cannot operate directly on arbitrary predicate logic sentences with nested connectives and quantifiers.

They require a flat, uniform structure (a set of clauses) so that resolution and unification can be applied mechanically.

Converting a natural language sentence into clausal form is therefore a seven-step pipeline, and every step exists to strip away one layer of complexity until only clauses remain.

Steps Involved

Step 1	Representing the Sentence in Predicate Logic
Step 2	Removing Implication (and Biconditional)
Step 3	Moving Negation Inwards
Step 4	Skolemization (Eliminating Existential Quantifiers)
Step 5	Moving Universal Quantifiers Outwards
Step 6	Converting to CNF
Step 7	Putting into Clauses

Step 2: Removing Implication (and Biconditional)

Clausal form only allows ¬, ∧, ∨ (plus quantifiers), no → or ↔.

Eliminate them using their logical equivalences.

$P \rightarrow Q \equiv \neg P \vee Q$

$P \leftrightarrow Q \equiv (\neg P \vee Q) \wedge (\neg Q \vee P)$

Step 2: Removing Implication (and Biconditional) …

$\forall x\,(Student(x) \land Studies(x) \rightarrow Passes(x))$

$\forall x\,(\neg(Student(x) \land Studies(x)) \lor Passes(x))$

Step 3: Moving Negation Inwards

Negations should apply only to atomic predicates, never to compound expressions or quantified formulas.

Use De Morgan's Laws and quantifier negation rules

Step 3: Moving Negation Inwards …

Double negation	$\neg\neg p \equiv p$
De Morgan's (AND)	$\neg(p \land q) \equiv \neg p \lor \neg q$
De Morgan's (OR)	$\neg(p \lor q) \equiv \neg p \land \neg q$
Negation of Universal Quantifier	$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
Negation of Existential Quantifier	$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$

Step 3: Moving Negation Inwards …

$\forall x\, (\neg(Student(x) \land Studies(x)) \lor P\ asses(x))$

$\forall x\, ((\neg Student(x) \lor \neg Studies(x)) \lor P\ asses(x))$

Step 4: Skolemization (Eliminating Existential Quantifiers)

Existential quantifiers cannot appear in clausal form.

They must be removed by replacing the existentially quantified variable with a witness.

- If it is not inside the scope of the for-all quantifier, replace with a brand-new Skolem constant (e.g.,c1).
- If it is inside the scope of universal quantifiers $\forall y_1, \ldots, \forall y_n$, replace with a new Skolem function $f(y_1, \ldots, y_n)$, since the witness may depend on those universally quantified variables.

Step 4: Skolemization (Eliminating Existential Quantifiers) …

Everyone has a mother

$\forall x\ (\text{Person}(x) \rightarrow \exists y\ \text{MotherOf}(y, x))$

$\forall x\ (\neg\text{Person}(x) \vee \exists y\ \text{MotherOf}(y, x))$

Since $\exists y$ is inside the scope of $\forall x$, Skolemize with a function $m(x)$ ("the mother of x"):

$\forall x\ (\neg\text{Person}(x) \vee \text{MotherOf}(m(x), x))$

Step 5: Moving Universal Quantifiers Outwards

After Skolemization, only universal quantifiers remain.

Move them all to the front of the formula.

$\forall x\ ((\neg Student(x) \lor \neg Studies(x)) \lor Passes(x))$

Step 6: Converting to CNF

Once the formula is quantifier-free (variables are implicitly universally quantified) and negation-normal

Use the distributive law to rearrange it into Conjunctive Normal Form

$$P \lor (Q \land R) \equiv (P \lor Q) \land (P \lor R)$$

$$\neg Student(x) \lor \neg Studies(x) \lor Passes(x)$$

Step 7: Putting into Clauses

Drop the universal quantifiers (all variables are now implicitly universally quantified)

Split the conjunction into a set of individual clauses.

This is the final clausal form used by resolution/Prolog.

$\{ \neg Student(x) \lor \neg Studies(x) \lor Passes(x) \}$

Example: Every student who does not study fails some exam.

Predicate logic	$\forall x\,(Student(x) \wedge \neg Studies(x) \rightarrow \exists y\,(Exam(y) \wedge Fails(x, y)))$
Remove implication	$\forall x\,(\neg(Student(x) \wedge \neg Studies(x)) \vee \exists y\,(Exam(y) \wedge Fails(x, y)))$
Negation inward	$\forall x\,((\neg Student(x) \vee Studies(x)) \vee \exists y\,(Exam(y) \wedge Fails(x, y)))$
Skolemization	$\forall x\,((\neg Student(x) \vee Studies(x)) \vee (Exam(e(x)) \wedge Fails(x, e(x))))$
Quantifiers outward	$\forall x\,((\neg Student(x) \vee Studies(x)) \vee (Exam(e(x)) \wedge Fails(x, e(x))))$
Converting CNF	$\forall x\,[(\neg Student(x) \vee Studies(x) \vee Exam(e(x))) \wedge (\neg Student(x) \vee Studies(x) \vee Fails(x, e(x)))]$
Clauses	$\{\neg Student(x) \vee Studies(x) \vee Exam(e(x))\}$ $\{\neg Student(x) \vee Studies(x) \vee Fails(x, e(x))\}$